

Week 2 – Quality Inspection System

1. Introduction

During Week 2 of the internship, an image-based **Quality Inspection System** was developed using Python and OpenCV. The main purpose of the system was to inspect product images and determine whether the product was in normal condition or had a visible defect.

The system processes different product images and performs basic computer vision operations such as image preprocessing, edge/contour analysis, and defect identification. Based on the inspection, each product is classified as **PASS** or **FAIL**. When a defect is detected, the affected area is highlighted in the output image.

The system was tested on multiple types of products, including a mouse, water bottle, a component with missing screws, and a broken watch. This provided practical experience in image processing and automated visual quality inspection.

2. Objective

The main objectives of this task were:

- To develop a basic automated quality inspection system.
- To process product images using OpenCV.
- To identify visible defects in products.
- To classify products as PASS or FAIL.
- To highlight detected defect areas.
- To save and review inspection results.

3. Tools and Technologies

Tool / Technology	Purpose
Python	Development of the inspection program
OpenCV	Image processing and defect detection
NumPy	Numerical and image data processing
Ubuntu / WSL2	Development environment

4. Working Methodology

The system follows a simple image-processing workflow. First, the product image is loaded into the Python program. The image is then processed using OpenCV techniques to identify important visual features.

Depending on the product, the system analyzes relevant regions and searches for abnormal features such as missing components or cracks. After the analysis, the system generates an inspection decision.

The general workflow is:

Input Image → Image Processing → Defect Detection → PASS/FAIL Decision → Result Visualization

Detected defects are marked on the output image so that the inspection result can be visually verified.

5. Test Results

The system was tested using four sample product images:

No.	Product	Condition	Detected Defect	Result
1	Mouse	Normal	None	PASS
2	Water Bottle	Normal	None	PASS
3	Component	Defective	2 Missing Screws	FAIL
4	Watch	Defective	Cracked Screen	FAIL

6. Conclusion

The Week 2 task successfully demonstrated a basic **computer vision-based quality inspection system**. The system was able to process multiple product images and classify the provided samples as PASS or FAIL. Defective samples such as the missing-screw component and cracked watch were identified and highlighted.

This task helped develop practical skills in **Python, OpenCV, image processing, contour/edge analysis, and automated quality inspection**. The system can be further improved in the future by using machine learning and real-time camera-based inspection.